

Knowledge Representation and Processing Models in Cognitive Psychology



Arndt Bröder, University of Mannheim



Lecture Series "Unwrapping the Mind"



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- Judgment and Decision Making
- Numerical estimation
- Metamemory and Memory
- Experiments & Formal Modeling

Agenda

1. Cognitive Psychology: Essentials

- Topic
- History
- Methods

Part 1

2. Representations and Processes

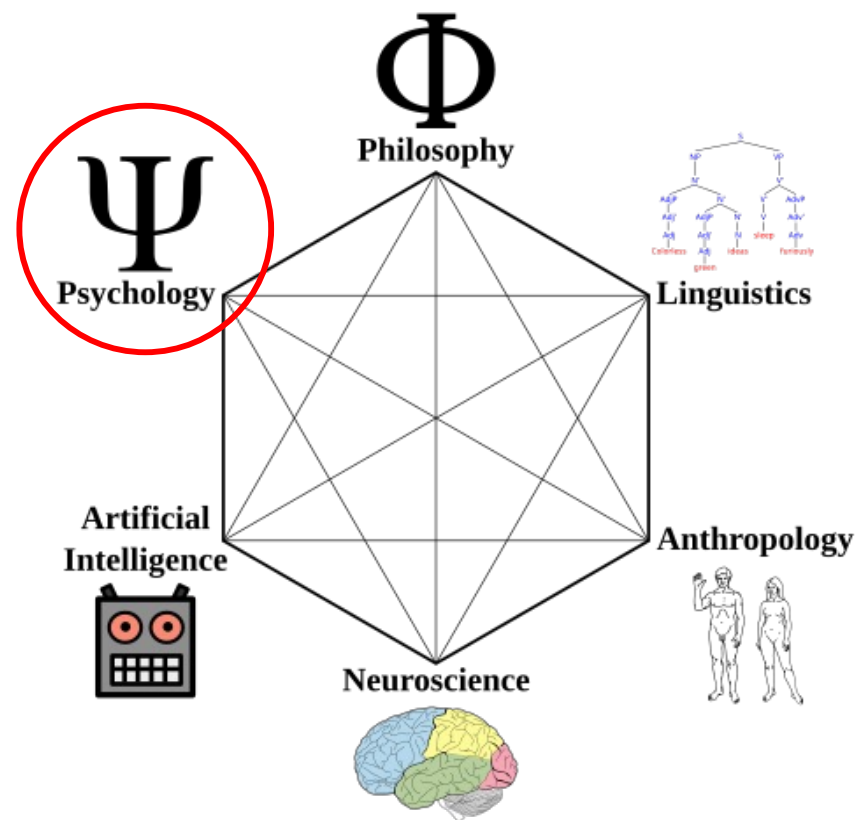
3. Cognitive Modeling of Numerical Estimation

- Artificial Flowers (Demonstration)
- Birds (Validation)

Part 2

4. Outlook

Part 1



1. What is Cognitive Psychology?

„Define Cognitive Psychology“



Cognitive psychology is a subfield of psychology that focuses on the scientific study of mental processes. These include perception, attention, memory, language, problem-solving, reasoning, and decision-making. It seeks to understand how people acquire, process, store, and retrieve information, often drawing on methods from experimental psychology, neuroscience, and computational modeling.

„Define Cognitive Psychology for a 12-year old“



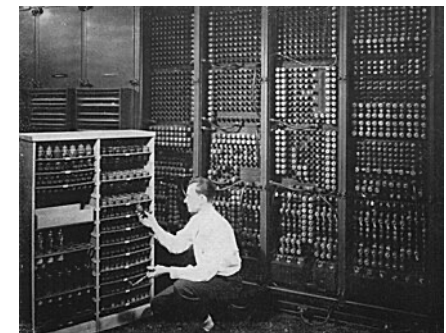
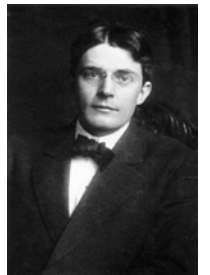
Cognitive psychology is the study of how our minds work—how we think, learn, remember things, pay attention, and solve problems.

It's like being a detective who tries to figure out what's going on inside people's heads when they read a book, do a puzzle, or try to remember someone's name. Cognitive psychologists study these things to understand how people think and why they sometimes make mistakes.

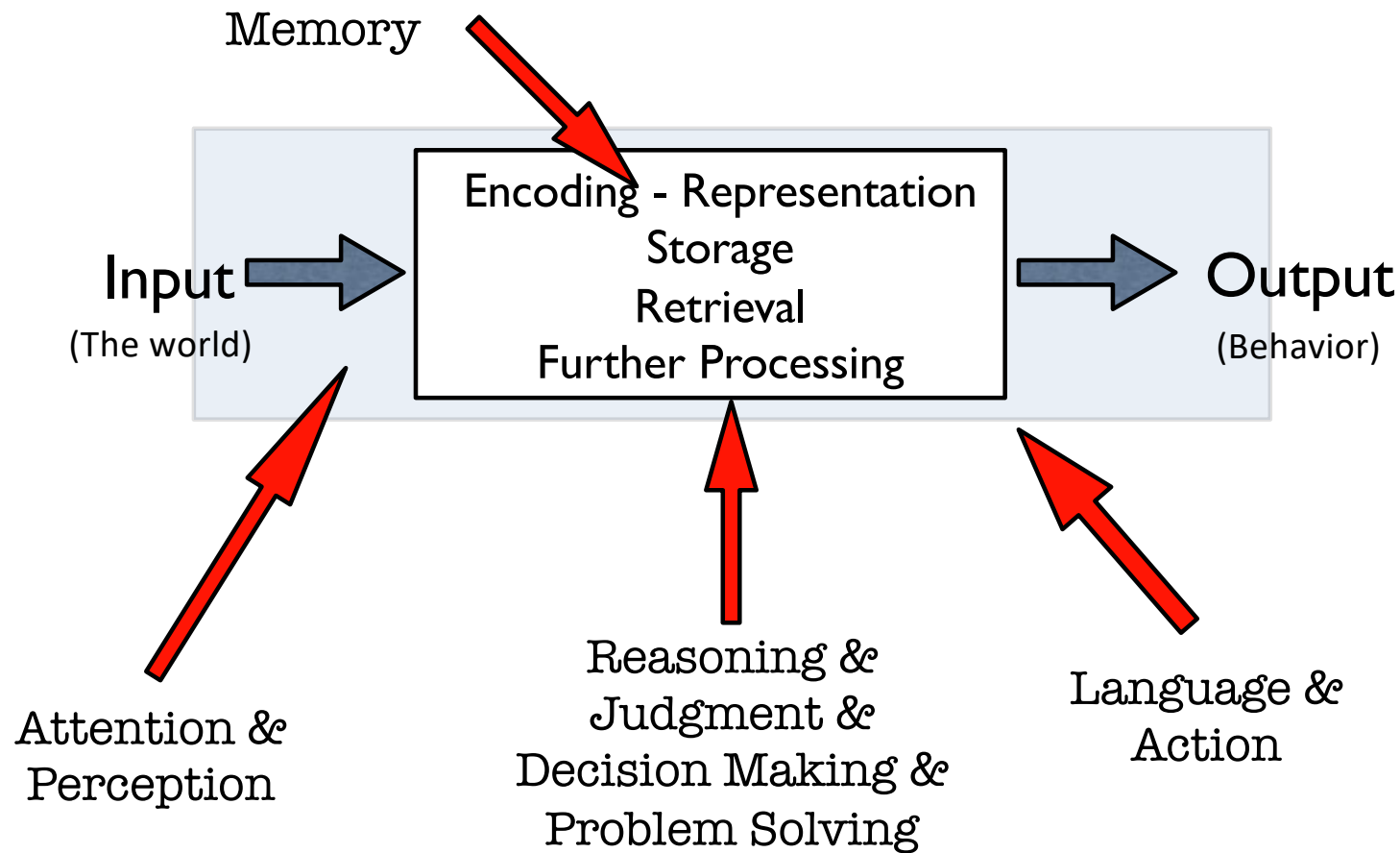
An ultra-brief history of Cognitive psychology



- 1879 – Wilhelm Wundt founds the first laboratory for Experimental Psychology in Leipzig
- 1885 – Hermann Ebbinghaus publishes his Habilitation *Über das Gedächtnis (On memory)*
- 1900s – Studies on thinking by the Würzburg School
- 1907 – Methodological controversy on introspection (Wundt vs. Bühler)
- 1913 – The „Behaviorist manifesto“ by J. B. Watson is published
- 1920s - The Gestalt psychologists work on creative problem solving
- 1920s to 1950s – Behaviorism reigns – no cognition allowed!
- 1956 – The cognitive revolution
- since 1990s – The neuroscientific revolution

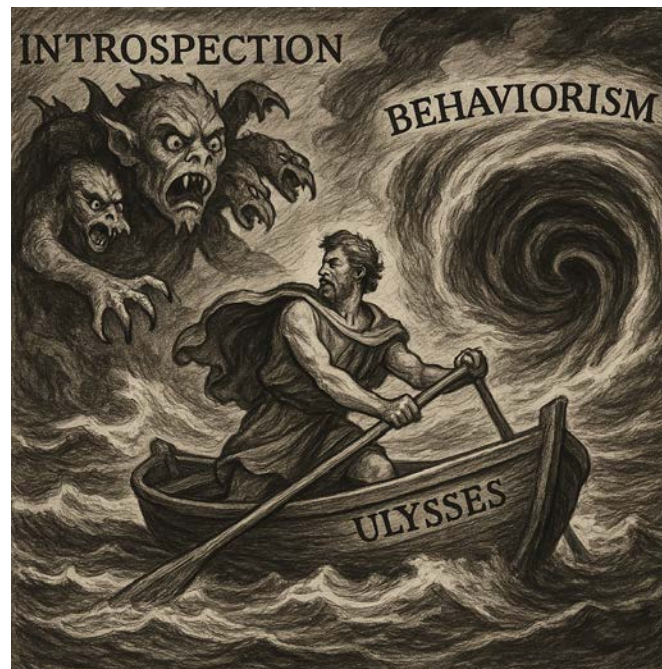


Information Processing as a Meta-Language



Methods of Cognitive Psychology

- How can we deal scientifically with immaterial and unobservable constructs?
- Are we caught between the Scylla of introspection and the Charybdis of Behaviorism?



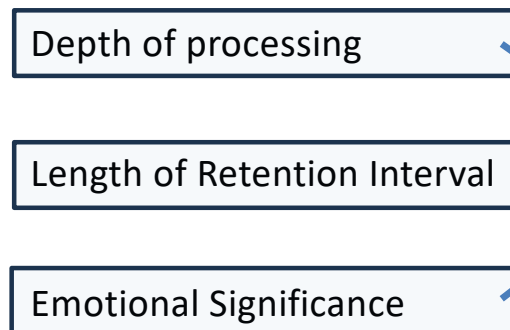
„Methodological Behaviorism“

Example: Memory for a List of Words

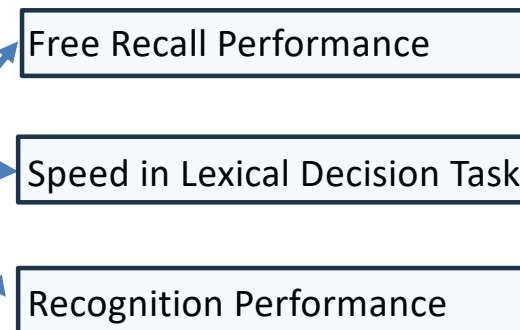


Edward C. Tolman
(1886-1959)

Independent variables:



Dependent Variables:

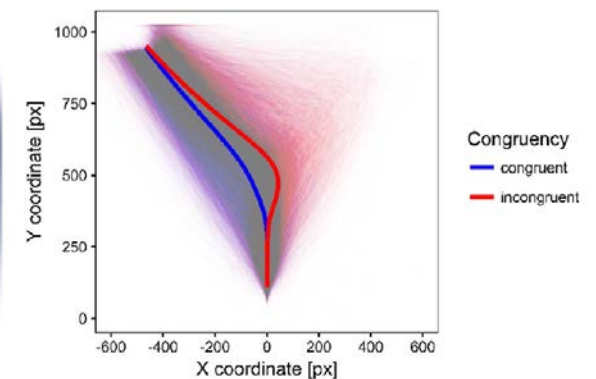
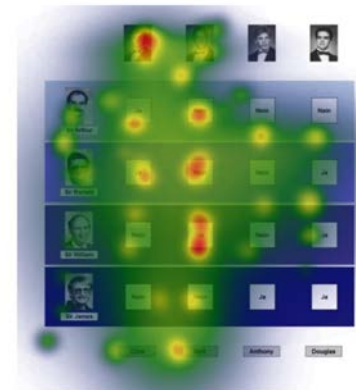
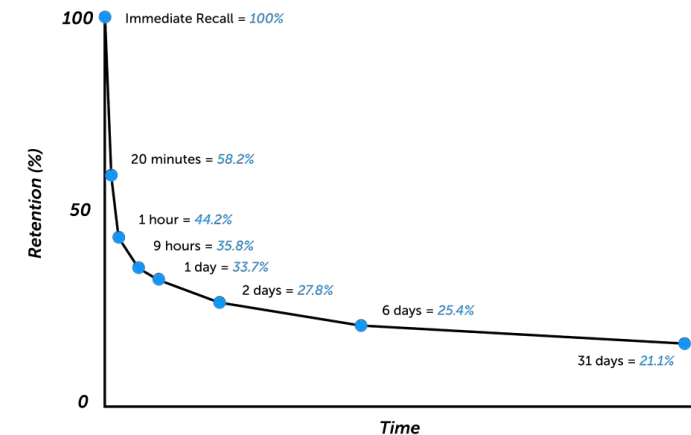


Strength of
Memory Trace

Ergo: Scientific theorizing about unobservable (theoretical) variables is possible strictly based on observables!

Methods in Cognitive Psychology

- General: Experiments
 - Variation of independent variables to assess causal effect on dependent variables
- Specific Methods:
 - solution rates (logical reasoning, problem solving)
 - reaction times
 - eye movements
 - mouse movements etc.
 - computational modeling



2. Ideas about Knowledge Representation



Kinds of knowledge representations



Feature Lists

Turkey:

- feathers
- beak
- big
- [can't fly]
- ...

Robin:

- feathers
- beak
- small
- can fly
- red breast
- ...

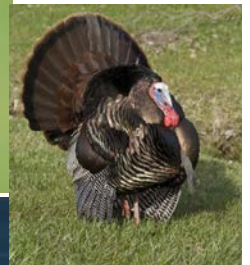
Blackbird:

- ...

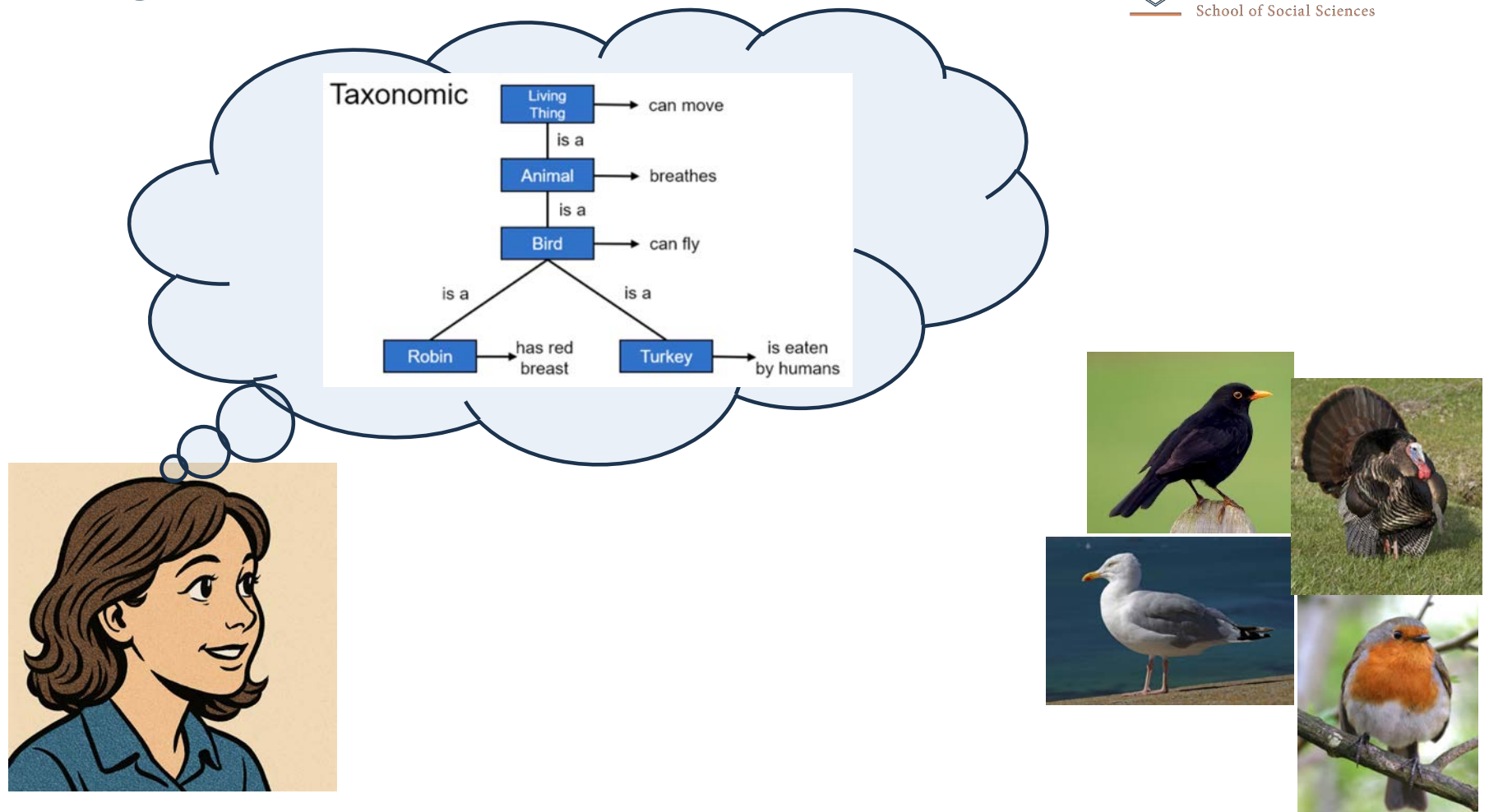


Useful for some tasks, but

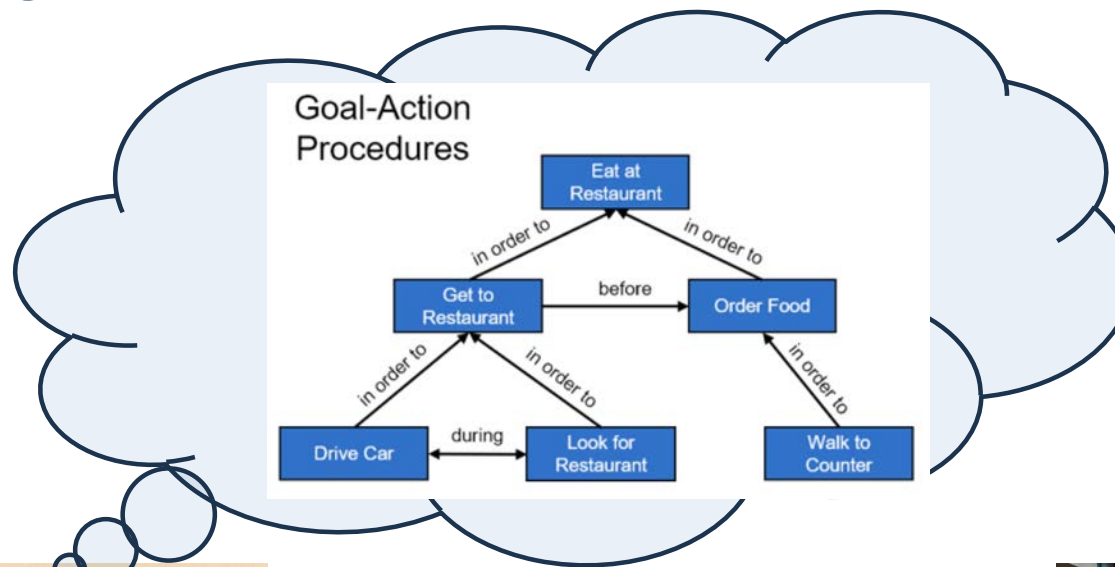
- simplistic
- no relations represented
- high storage demand



Knowledge structures: Semantic Networks



Knowledge structures: Goal-Action-Procedures

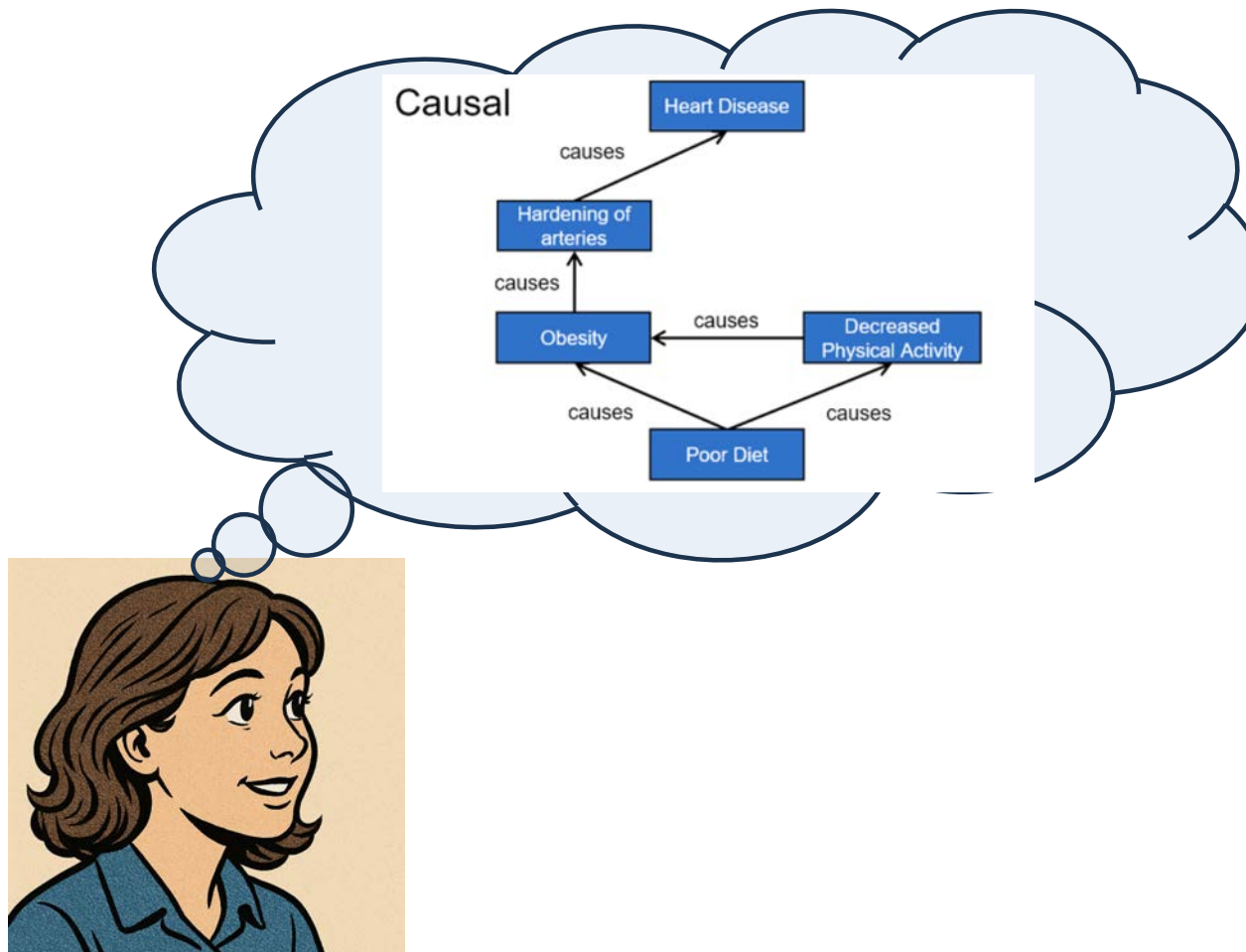


Also:

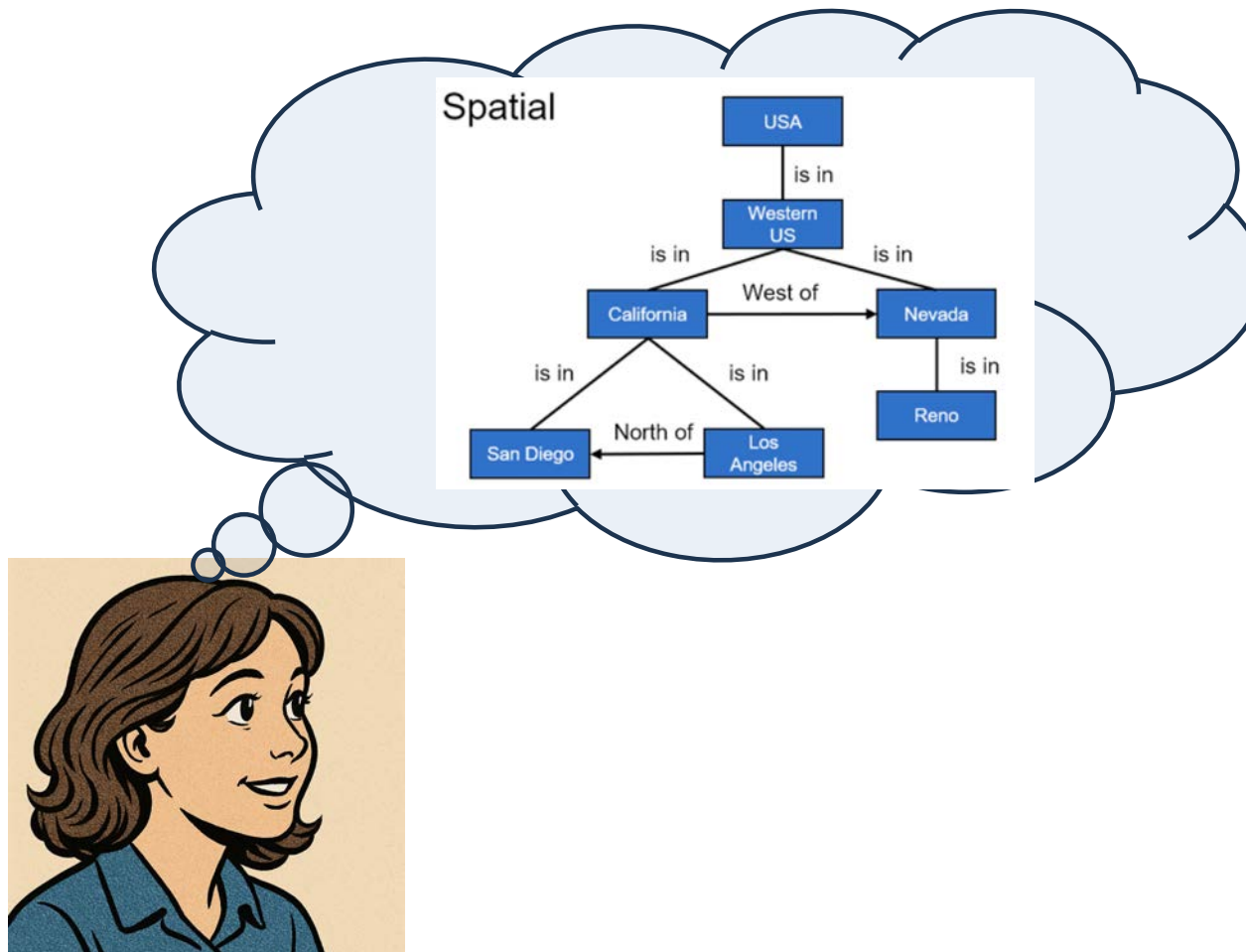
Scripts containing a typical chronological sequence of events
Slots are filled with defaults that need not be represented explicitly in memory



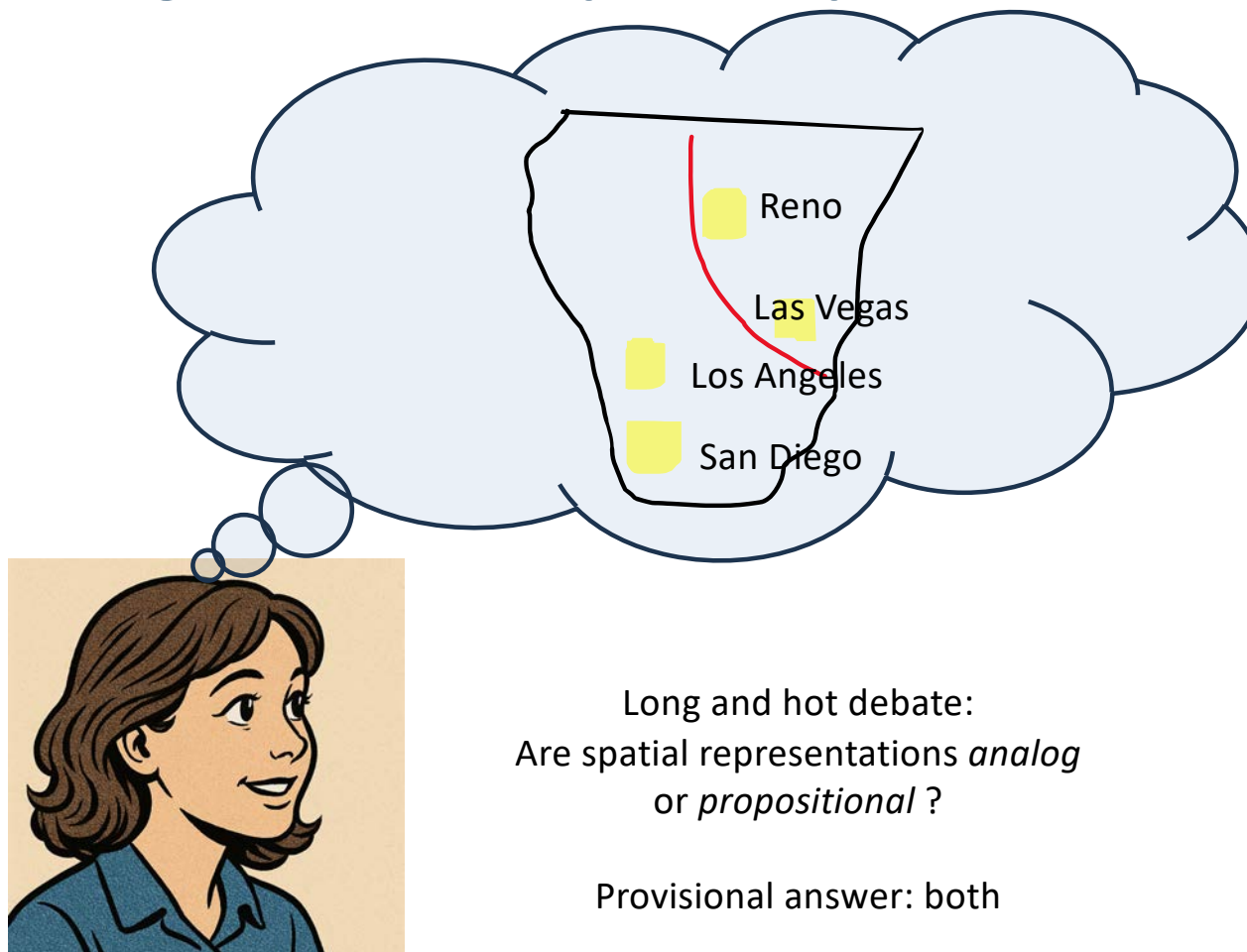
Knowledge structures: Causal Structures



Knowledge structures: Spatial Representations



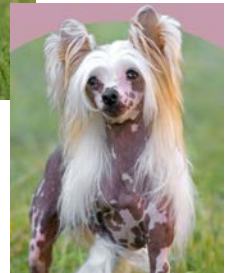
Knowledge structures: Spatial Representations



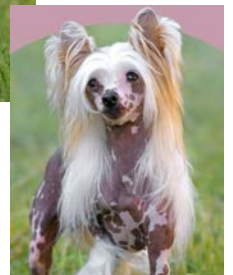
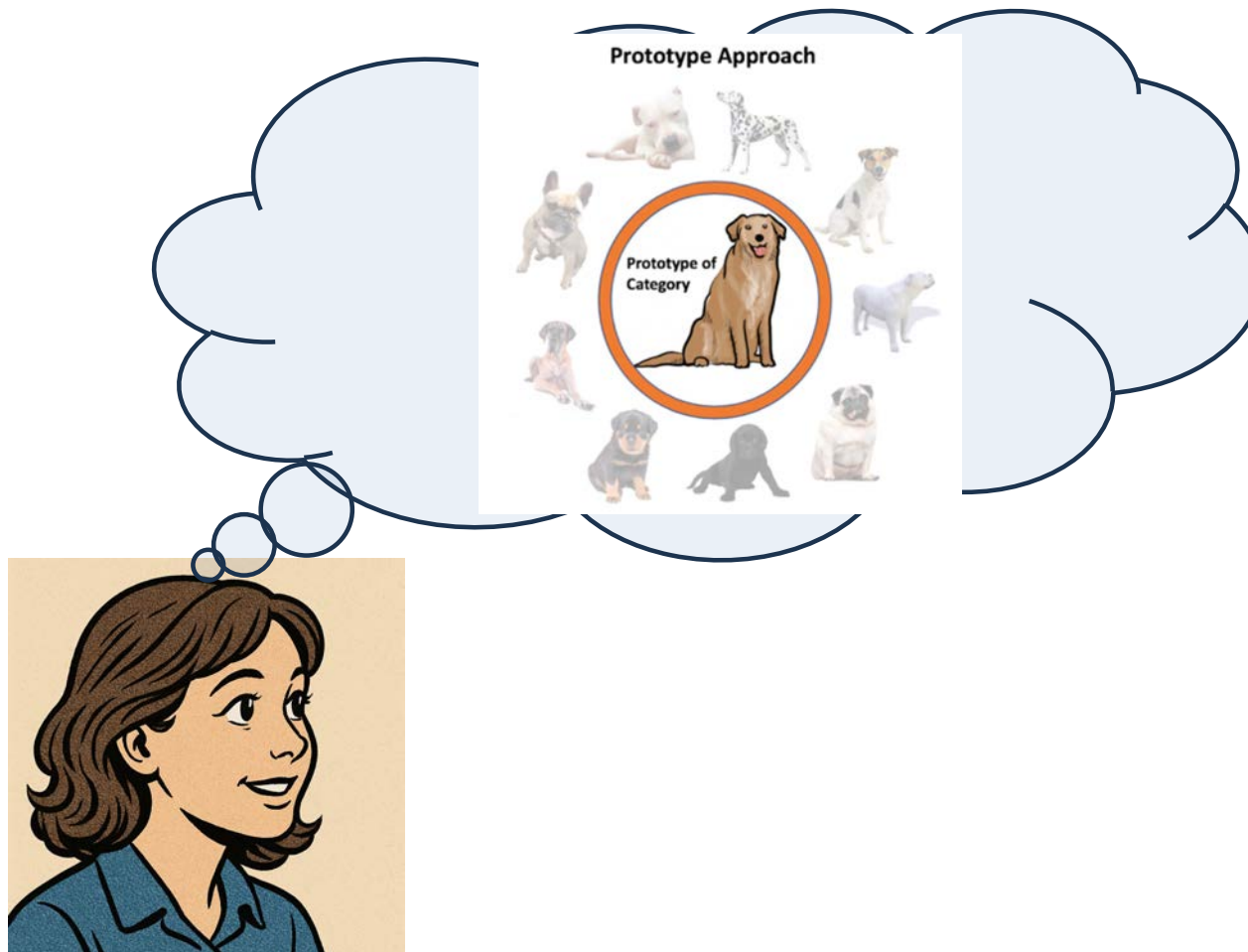
Similarity-based category-structures

„Similarity plays a crucial role in making predictions because similar things usually behave similarly.“

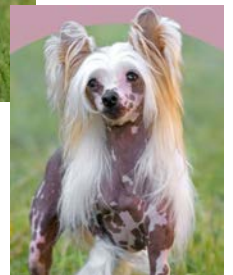
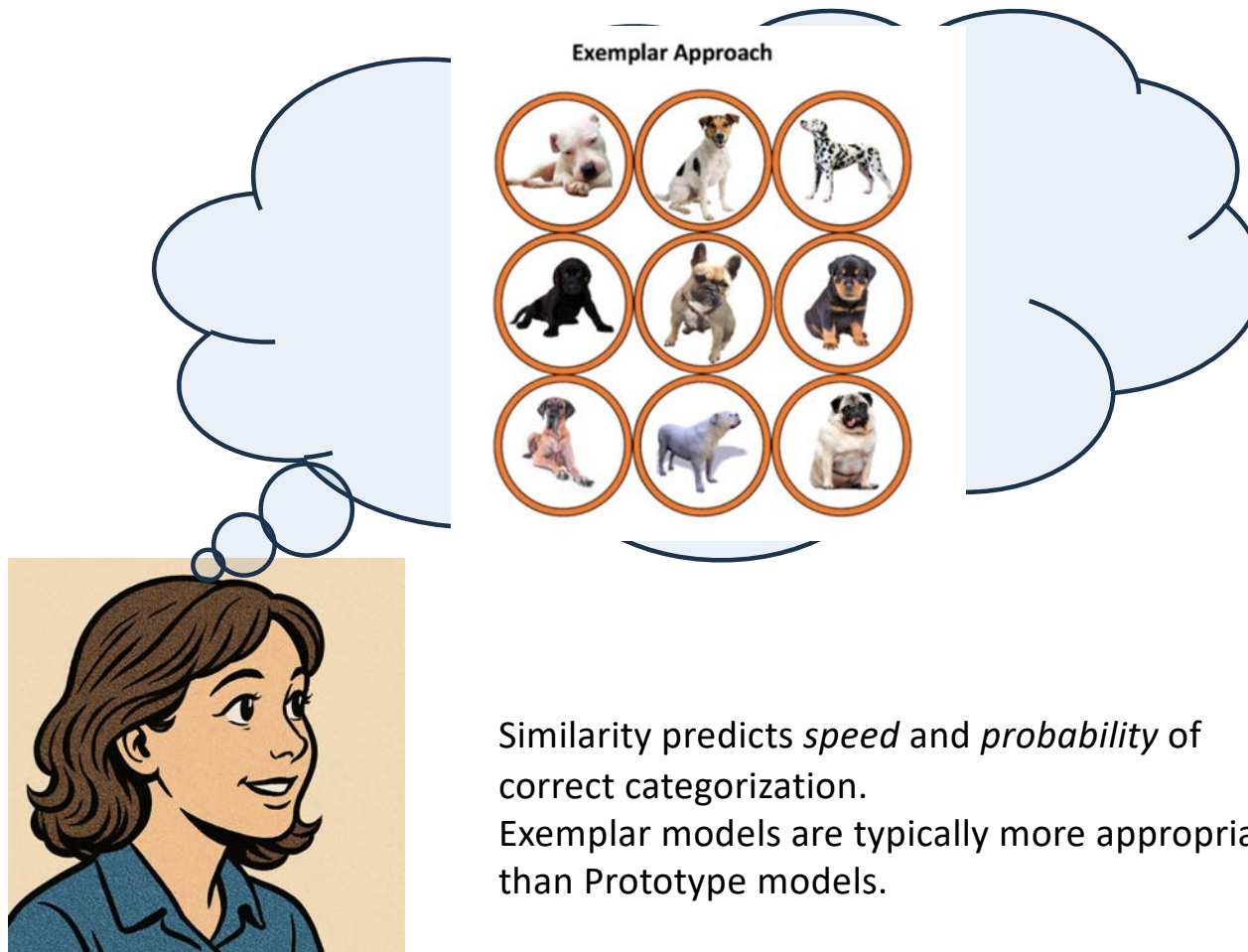
(Goldstone & Son, 2005)



Similarity-based category-structures



Similarity-based category-structures



Summary on Representations and Processes



- Representations are abstracted models of the world (with gaps)
- Certain representations are adequate for certain processes
- There is evidence for multiple representational systems
 - **analog** → explains reaction times in mental rotation or map scanning
 - **propositional** → explains logical abilities
 - **semantic networks** → explain reaction times in verification tasks and semantic priming
 - **perceptual similarity** → explains reaction times and errors in classification tasks
 - **features** → explain strategies in judgment and decision tasks

Summary on Representations and Processes



- „The mind is likely to make use of a **variety of forms of representation**. We have a large number of cognitive mechanisms that have evolved over millions of years to accommodate a number of **disparate cognitive abilities**. There is no reason to think that the representations (...) that are best for understanding low-level vision or basic motor control are also (...) appropriate for handling the kinds of (...) structures that people encounter when reading academic prose.“ (Markman, 2012, p. 49)

Part 2: Representation and Processes in Estimation



What Influences Our Judgment?

9 October 2023

The German Research Foundation has awarded funding in the amount of almost 362,000 euros to Dr. David Izydorczyk of the Chair of Experimental Psychology to investigate this question.



How long will it take me to get to the office? How many calories does this meal have? What is the carbon footprint of this product? Estimating quantities or numbers is part of our daily life and the subjective answers to these questions influence our behavior considerably. Thus, the exactness of our estimates is important. For research on this topic, the German Research Foundation (DFG) has awarded funding in the amount of almost 362,000 euros to Mannheim psychologist Dr. David Izydorczyk of the Chair of Experimental Psychology.

In research, there are some mathematical models which try to show and describe judgment and estimation processes. However, these models have only been tested in lab settings. Another research field uses realistic tasks – for example estimating the sugar content of food or the carbon footprint of products and investigates simple training methods which should help to improve the estimates people give. However, theories in this field are formulated imprecisely.

With this project, Izydorczyk wants to bring together both perspectives. The researcher hopes to get more insights into

Part 2: Representation and Processes in Estimation

- Estimation important everyday activity
 - How many calories in this dessert?
 - How long will it take to reach the main station?
 - Will my cash suffice for the groceries?
 - ... etc...
- State of the art:
 - Everyday tasks – no cognitive modeling, yet
 - Cognitive modeling – restricted to artificial and constructed stimuli
 - How can we model estimation with realistic stimuli?



Quantitative Judgment Processes

Rule-based processes

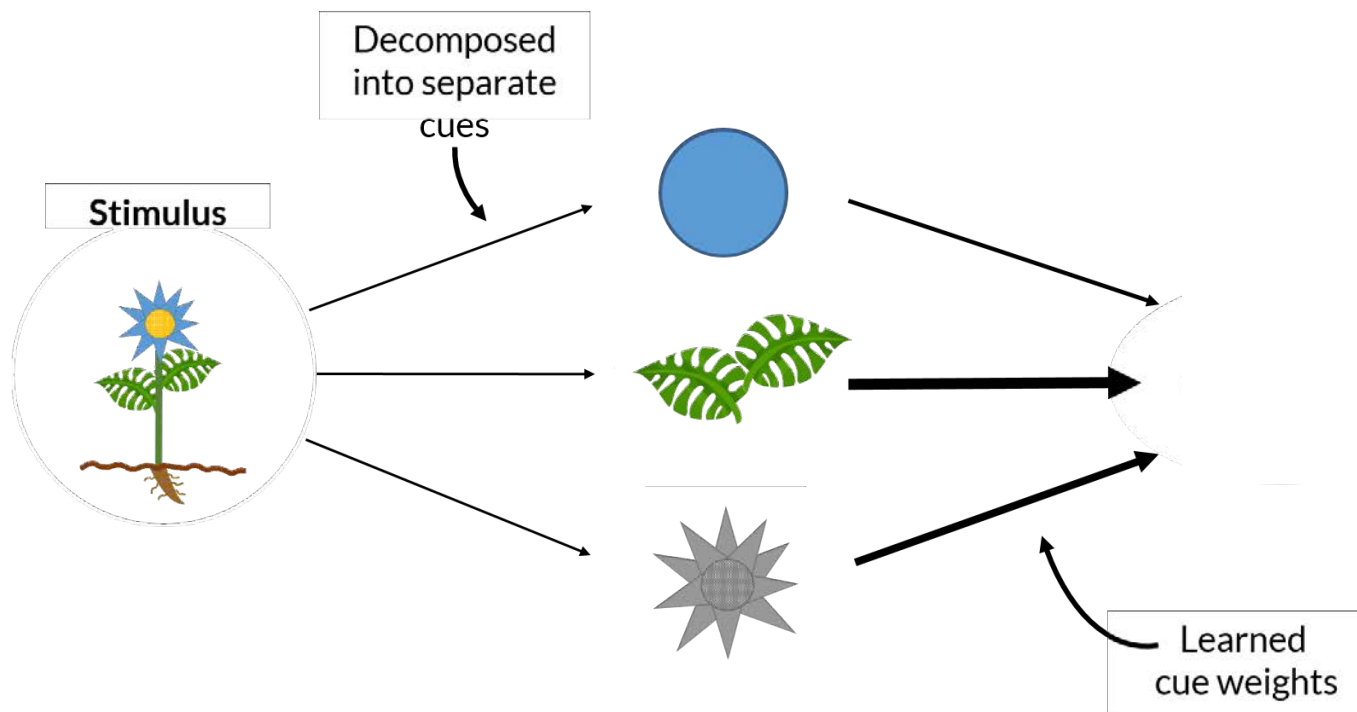
(e.g., Brehmer, 1994)



Quantitative Judgment Processes

Rule-based processes

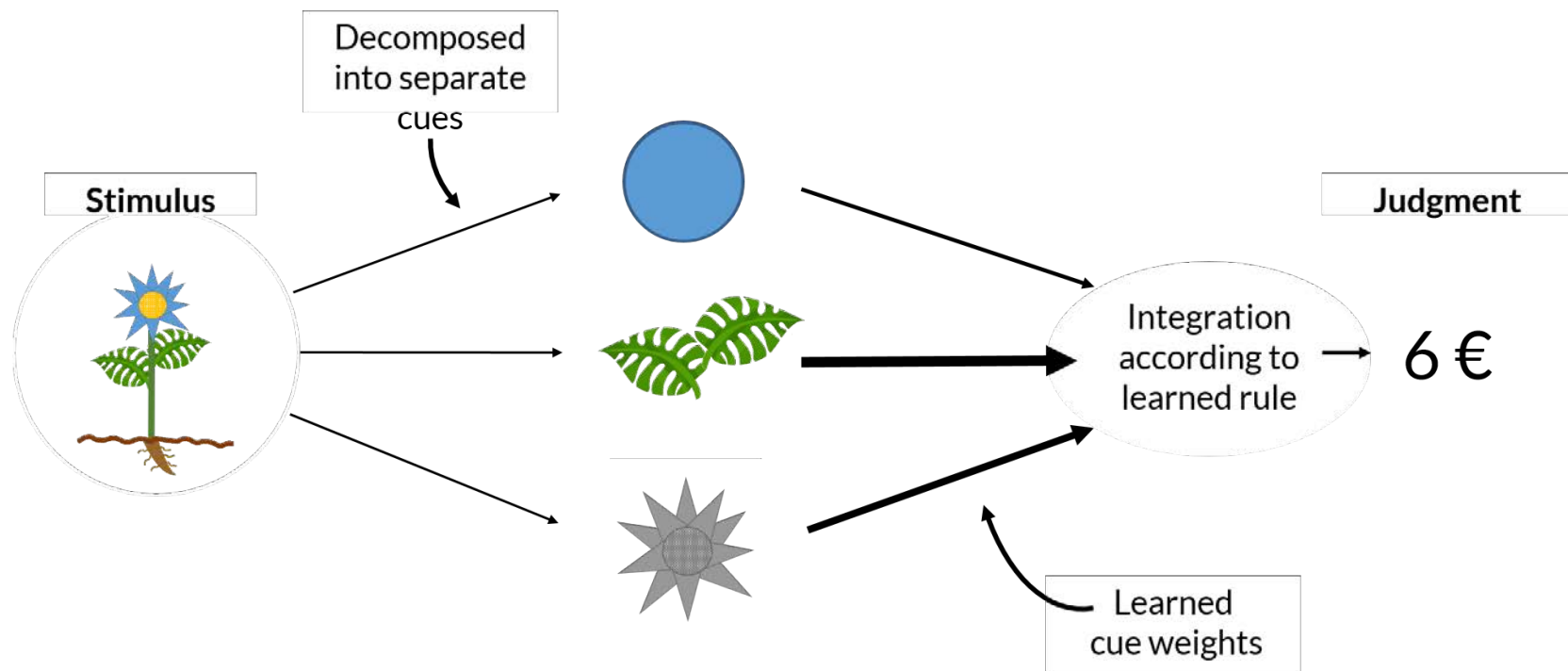
(e.g., Juslin et al., 2003)



Quantitative Judgment Processes

Rule-based processes

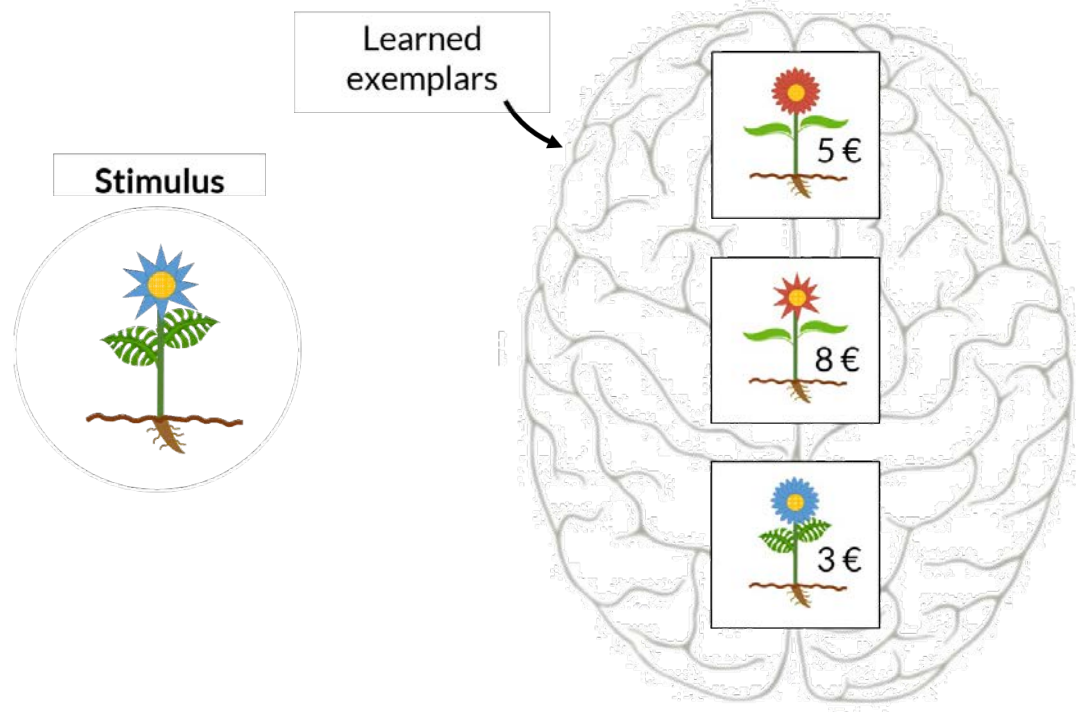
(e.g., Juslin et al., 2003)



Quantitative Judgment Processes

Exemplar-based processes

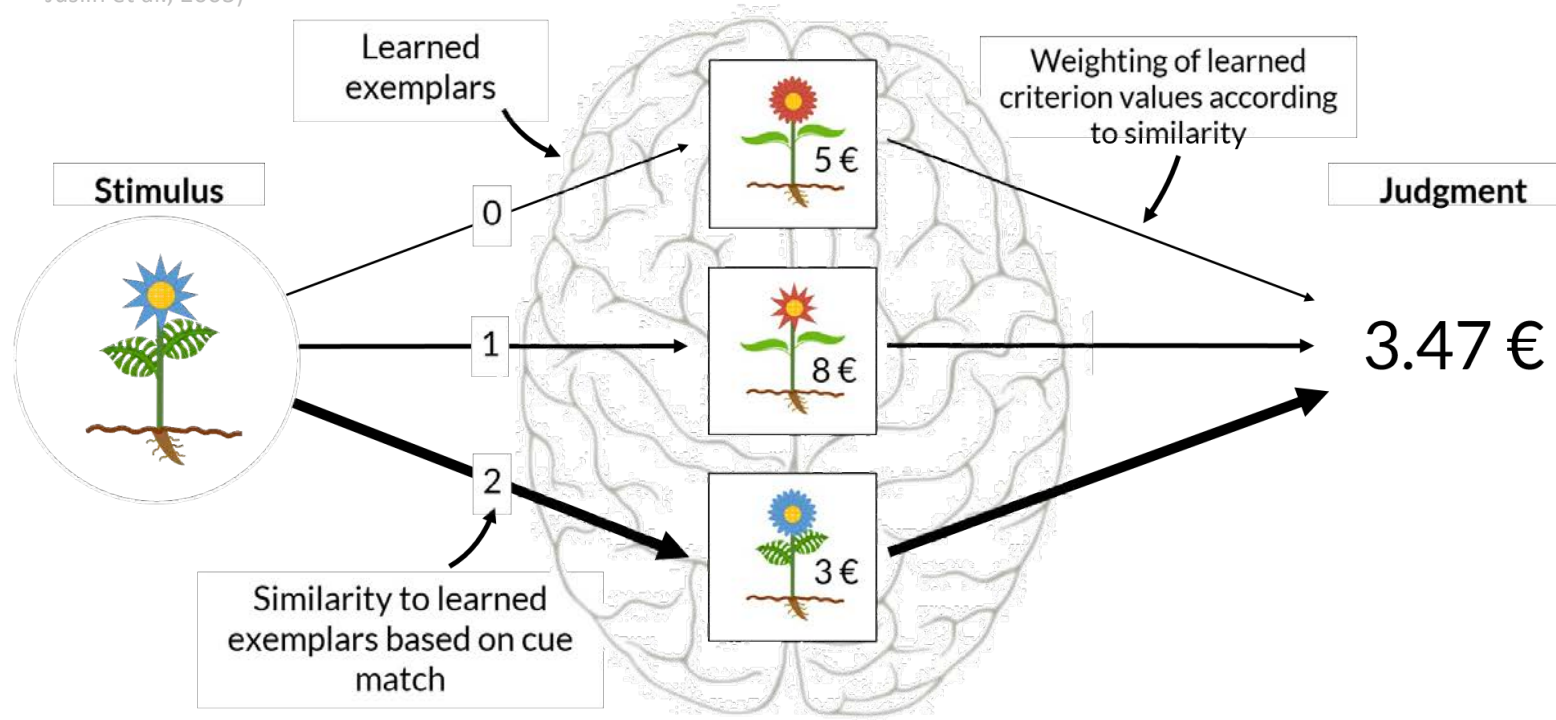
(Medin and Schaffer, 1978; Nosofsky, 1984;
Juslin et al., 2003)



Quantitative Judgment Processes

Exemplar-based processes

(Medin and Schaffer, 1978; Nosofsky, 1984;
Juslin et al., 2003)



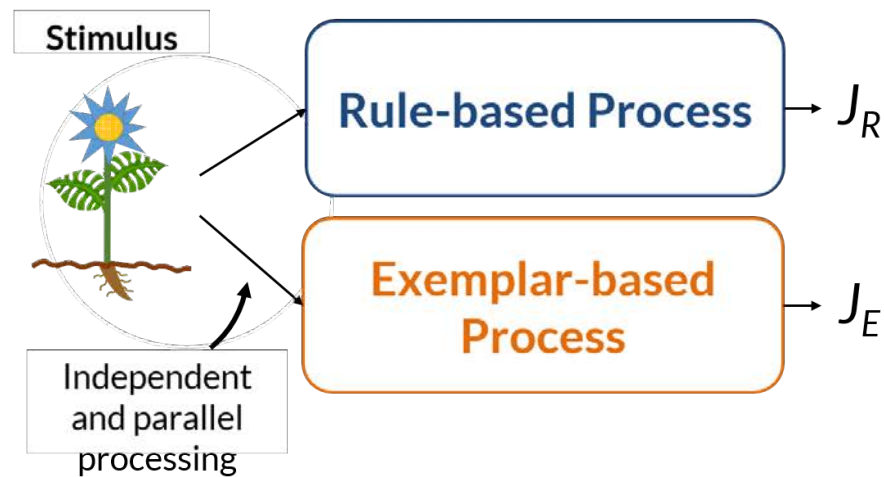
Quantitative Judgment Processes

A combination of both

(Albrecht et al. 2020; Bröder et al. 2017)

RulEx-J Model

(Bröder et al., 2017)

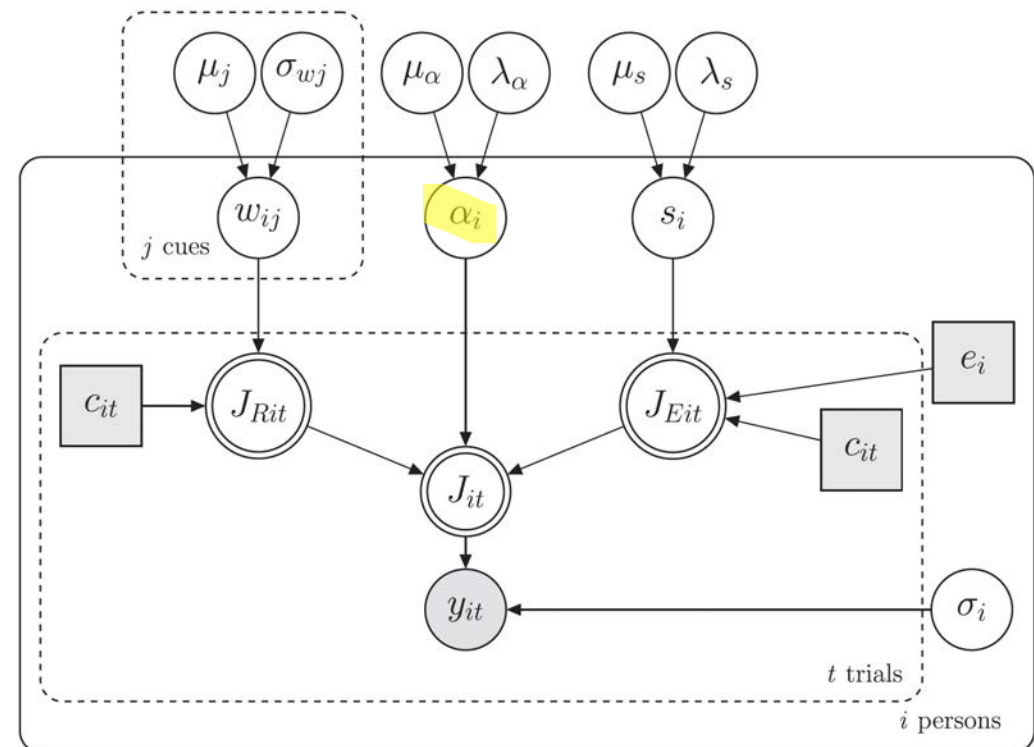


Further development

- Hierarchical Bayesian implementation of RulEx-J to deal with statistical issues (e.g., greater flexibility of rule module; Izydorczyk & Bröder, 2022)

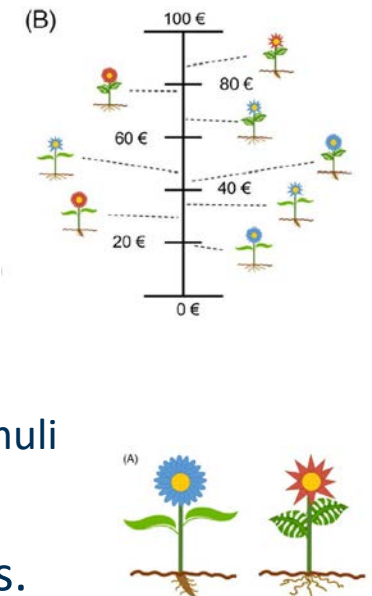
Figure 1

Graphical Model Representation of the Hierarchical Bayesian RulEx-J Model



The typical procedure for testing exemplar models

- **Training** phase with feedback
 - Participants estimate criterion for various *artificial* stimuli (repeatedly) with feedback = exemplars
- **Test** phase without feedback
 - Participants estimate criterion for old stimuli (exemplars) and new *transfer* stimuli
- Analysis: Assess best-fitting model based on the stimuli's cue patterns.
- Result: RuEx-J (52% best fit) superior to both exemplar model (19%) and rule model (29%) alone



Summary part 1 (The Lab)



- Sophisticated computational theories exist for modeling and explaining quantitative judgments
 - exemplars (Juslin et al., 2003; Karlsson et al., 2008)
 - rules (Brehmer, 1994; Brunswik, 1952; von Helversen & Rieskamp, 2008)
 - combinations (Albrecht et al., 2020; Bröder et al., 2017)
- Have been tested with ***highly controlled (artificial) stimuli*** in the lab
- Reason: Computational models need defined (and diagnostic) cue patterns as input

3. Modeling Judgment with realistic stimuli

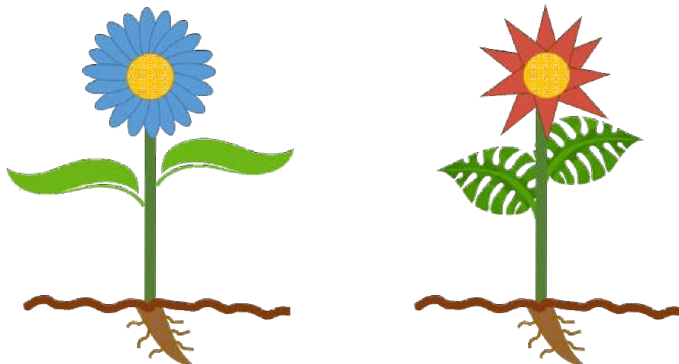
Goldstein and Hogarth (1997, p.37): *“To what extent can we generalize from laboratory studies with abstract tasks [and artificial stimuli] to behavior in the real-world domains?”*

- **Goal:** extract psychologically functional cues from naturalistic stimuli
- **Approach:** Extract cues from similarity ratings with multidimensional scaling (MDS)
- Borrowed from categorization research: Nosofsky et al. (2018a, 2018b, 2018c, 2019)



Two Studies with the same general procedure

Study 1: Validation
with artificial stimuli &
data from Izydorczyk & Bröder (2022):



Study 2:
with complex natural stimuli



Multidimensional Scaling

(1) Pairwise similarity ratings



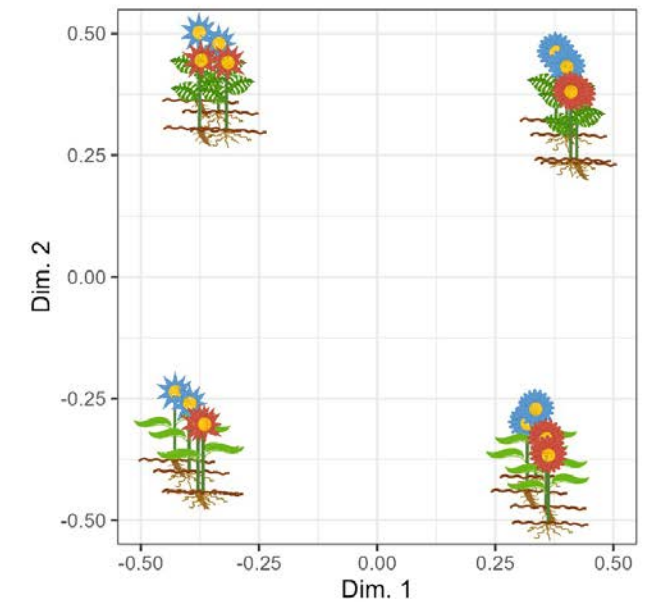
How similar are these two flowers?

not similar

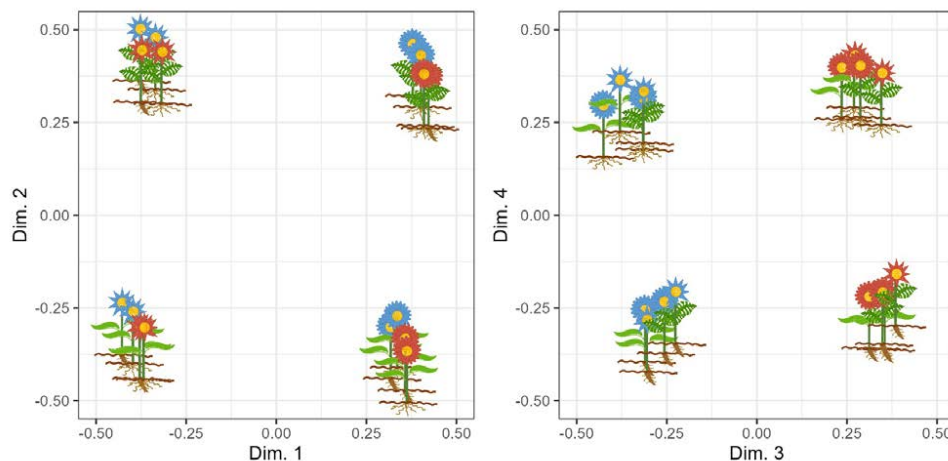
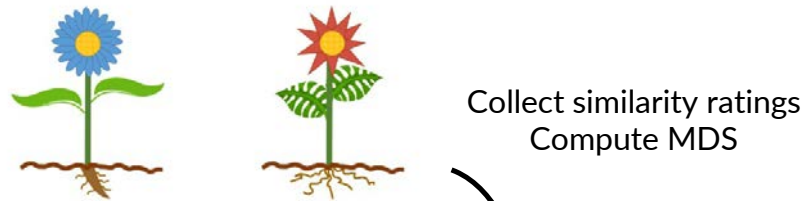
1234567

very similar

Respond by pressing the corresponding button on your keyboard



Study 1 Flowers: Validation

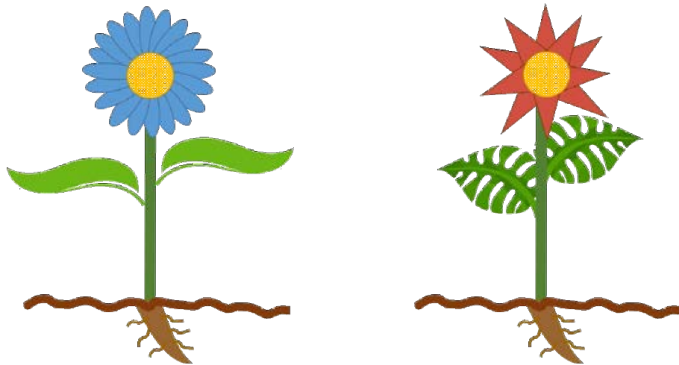


Results MDS

- Best MDS solution has 4 dimension (= # of features stimuli actually have)
- High correlation between observed and predicted pairwise-similarities ($r(118) = .99, p < .001$)

Study 1: Validation

with artificial stimuli &
data from Izydorczyk & Bröder (2022):



- Perfect recovery of cues using MDS analysis
- Same results when using MDS cues instead of exp. defined cues



Study 2:

with complex stimuli



- **Experiment:** Replication of Pachur & Olsson (2012) and Trippas & Pachur (2019)
- **General finding:** More rule-based processing when trained with *learning by comparison* than with *direct criterion learning*

General Procedure

Pairwise Similarity Ratings of $K = 32$ Birds ($N = 101$)

Two square boxes side-by-side. The left box contains a duck with a black head, orange beak, and brown and white body. The right box contains a small brown and white sparrow.

How similar are these two birds?

not similar

very similar

1	2	3	4	5	6	7
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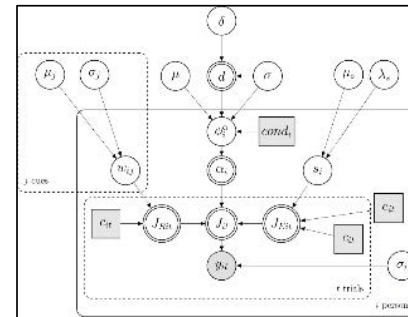
Respond by pressing the corresponding button on your keyboard

Every participant rated 124 out of all 496 possible pairs

MDS Analysis

Bird	Dim. 1	Dim. 2	Dim. 3
1	0.1	-0.2	0.3
2	0.2	-0.3	0.4
...			
32	1.4	-0.1	1.2

Computational Model



as cues in

to model/analyze

Judgment Experiment

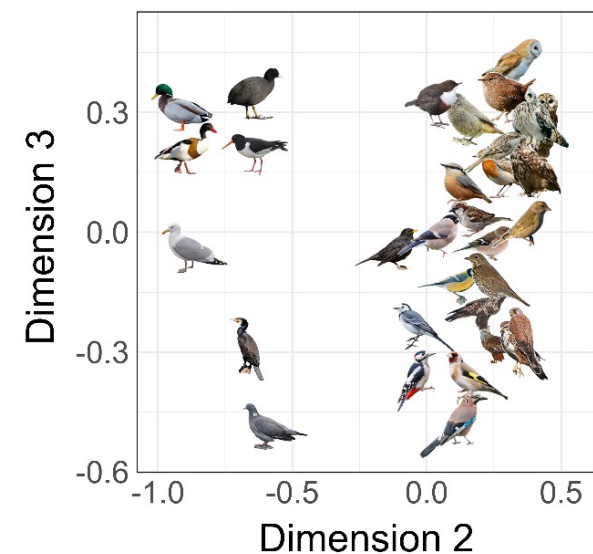
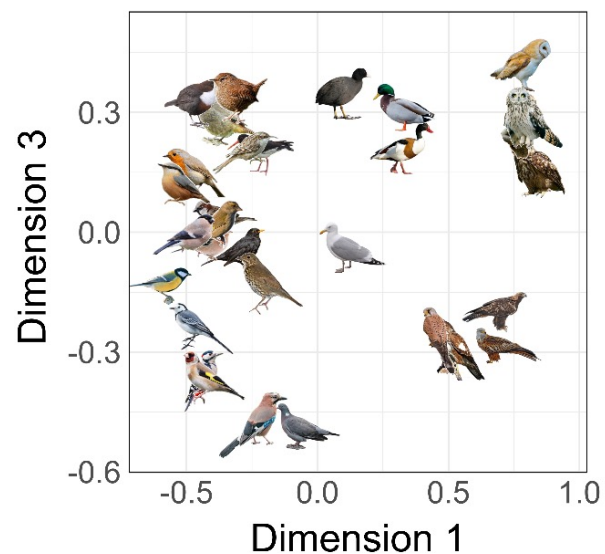
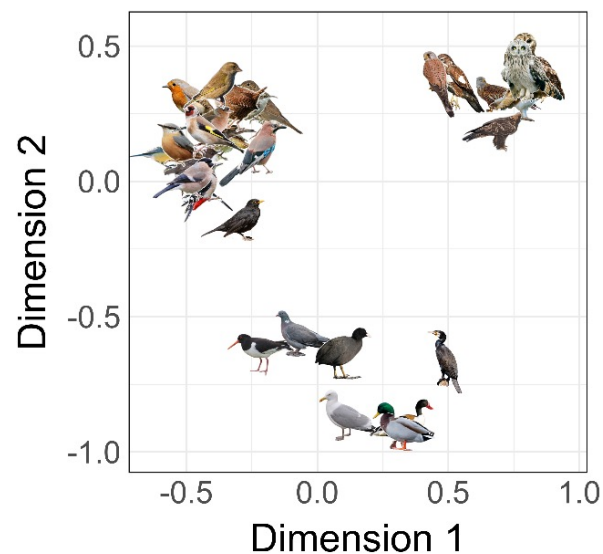


What is the top speed of this bird ?

Study 2: Birds

Results MDS

- Best MDS solution has **3 dimensions**
- High correlation between observed and predicted pairwise-similarities ($r(494) = .93, p < .001$)

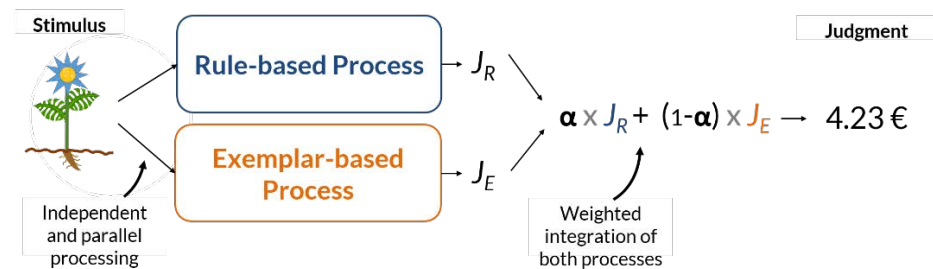


Main Hypothesis

based on Pachur & Olsson (2012) and Trippas & Pachur (2019):

More rule-based processing when trained with *learning by comparison* than with *direct criterion learning*

Data analyzed with
RulEx-J Model
(Bröder et al., 2017;
Izidorczyk & Bröder, 2022)



$$\bar{\alpha}_{\text{learning by comparison}} > \bar{\alpha}_{\text{direct criterion learning}}$$

Procedure: Judgment Experiment

Training Phase:

$k = 12$ exemplar birds

learning by comparison
($N = 39$)

direct criterion learning
($N = 39$)



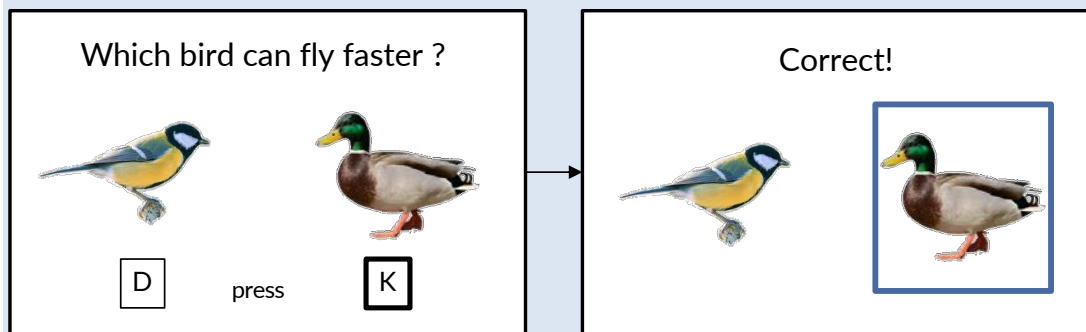
Testing Phase:

$K = 32$ birds (12 old, 20 new)

Procedure: Judgment Experiment

Training Phase:

$k = 12$ exemplar birds



learning by comparison
($N = 39$)

direct criterion learning
($N = 39$)

Testing Phase:

$K = 32$ birds (12 old, 20 new)

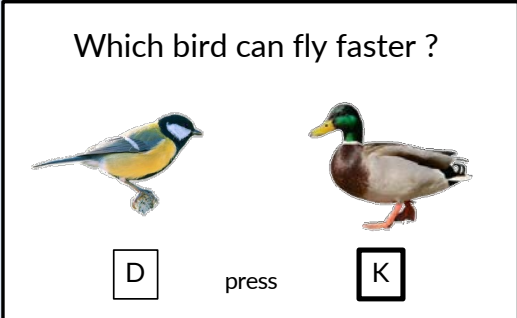
Procedure: Judgment Experiment

Training Phase:

$k = 12$ exemplar birds

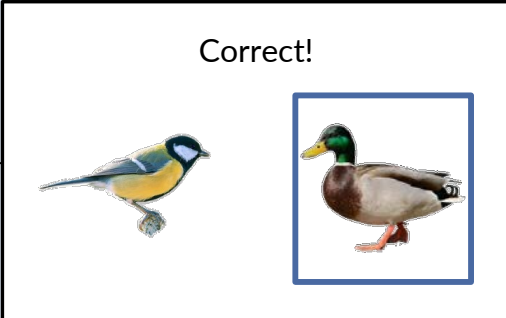
learning by comparison
($N = 39$)

Which bird can fly faster ?



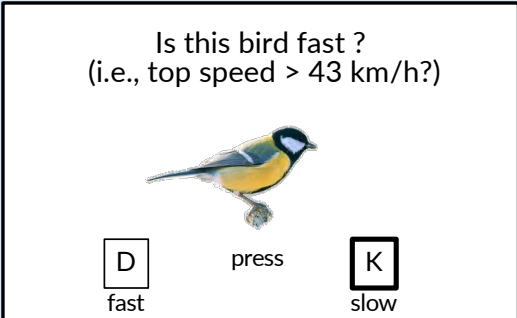
D press K

Correct!



direct criterion learning
($N = 39$)

Is this bird fast ?
(i.e., top speed > 43 km/h?)



D fast press K slow

Correct!



This bird is slow:
Top speed = 35 km/h

Testing Phase:

$K = 32$ birds (12 old, 20 new)

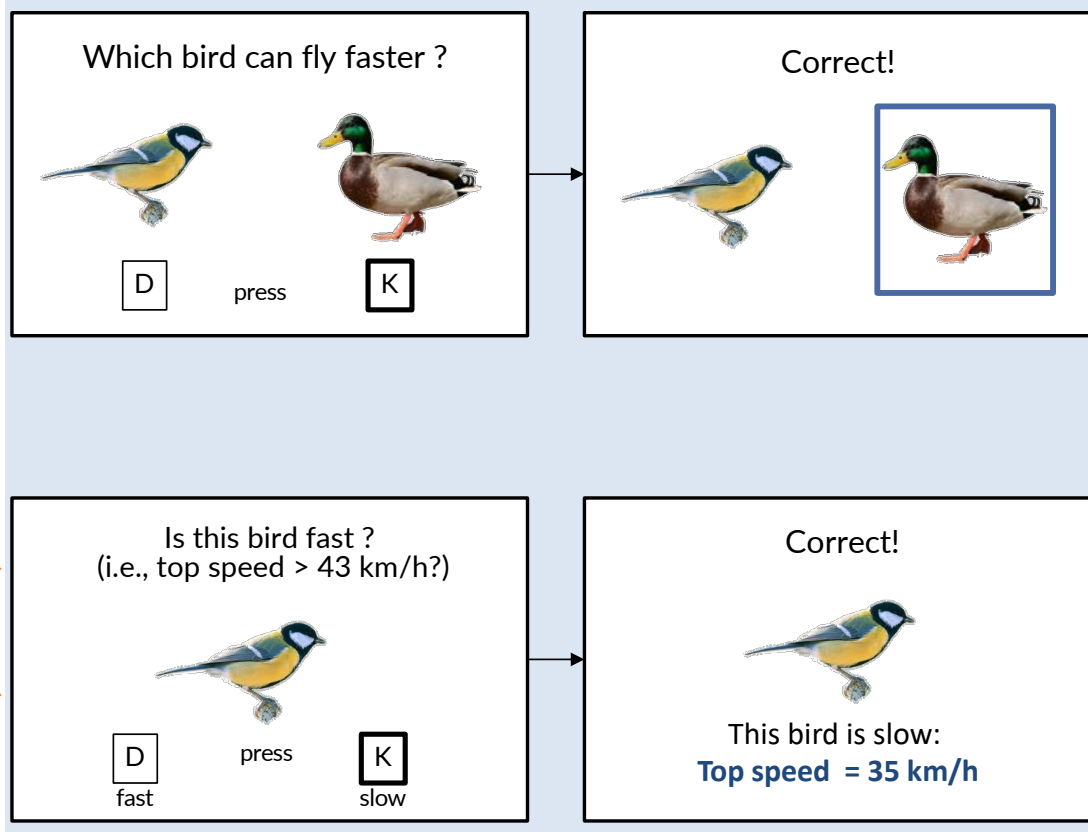
Procedure: Judgment Experiment

Training Phase:

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direct criterion learning
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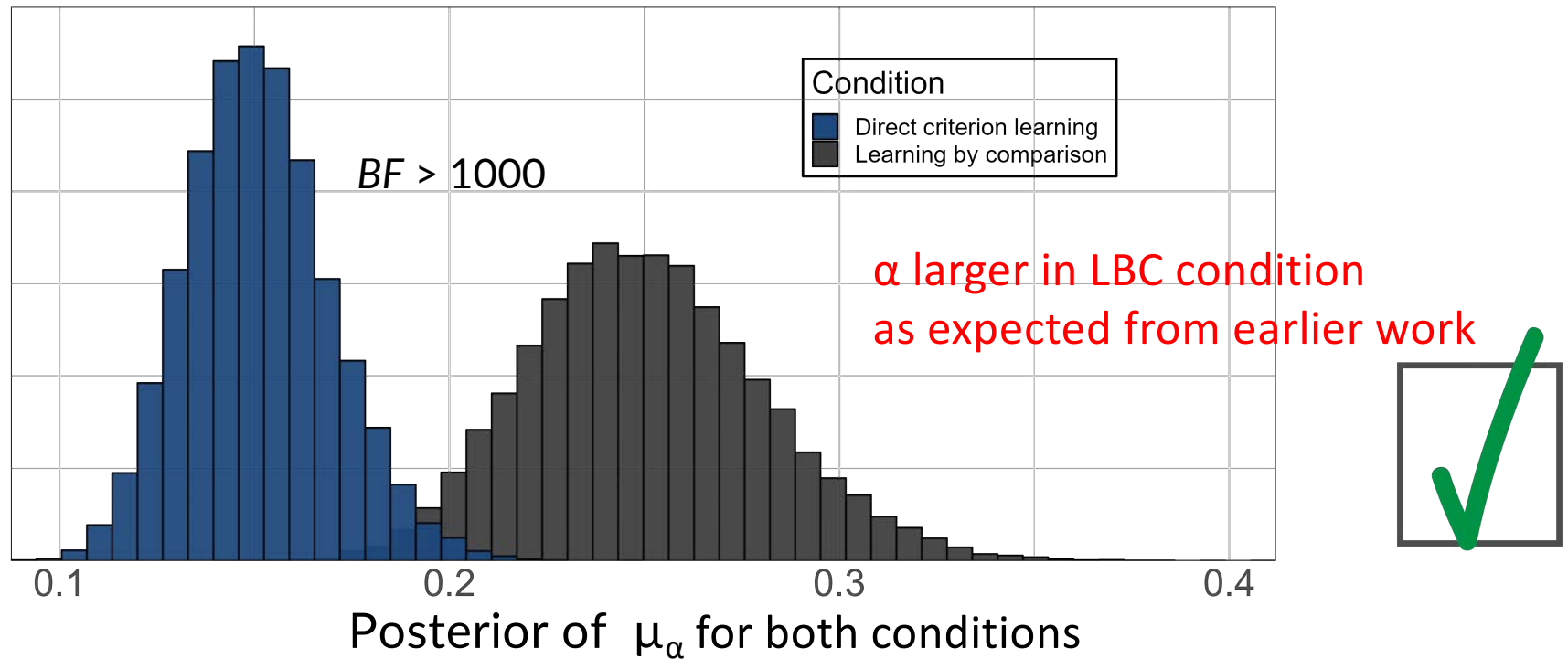


Testing Phase:

$K = 32$ birds (12 old, 20 new)



Results – Cognitive Modeling



$\alpha = 1$: Only rule-based processing

$\alpha = 0$: Only exemplar-based processing

Discussion & Open Questions

Goldstein and Hogarth (1997, p.37): *“To what extent can we generalize from laboratory studies with abstract tasks [and artificial stimuli] to behavior in the real-world domains?”*

Encouraging result:

- Modeling with MDS-extracted cues/features of naturalistic stimuli replicated a well-documented finding achieved with more controlled stimuli (death bugs)
- Method feasible for cognitive modeling of other domains?

Project Plans

- Generate Databases of similarity ratings and MDS-coordinates
 - Mammals
 - Food items
 - Countries

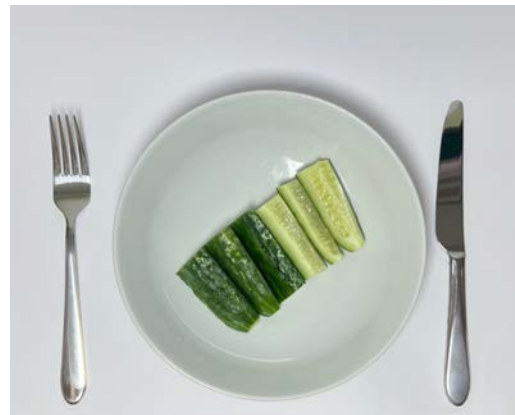
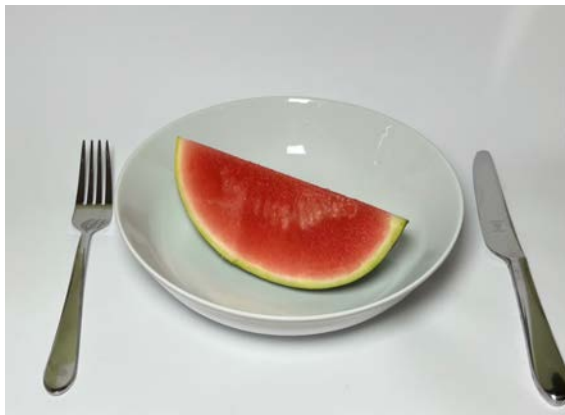
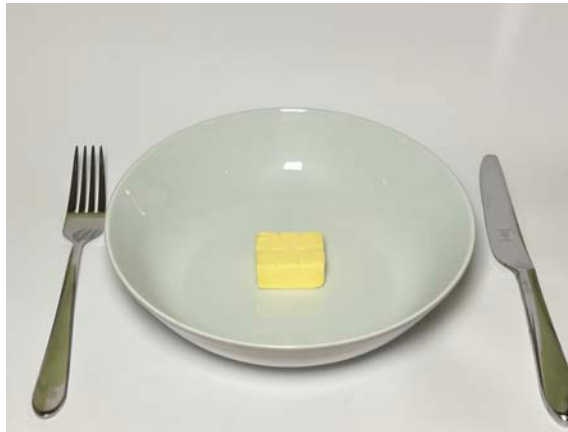
} multiple judgment criteria possible
- Model various seeding experiments with various computational models
 - Rule, Exemplar, RulEx-J, **Mapping Model, QuickEst, Guessing**
 - Moderators of mechanism use?

Example Mamals

80 mammals =
3,160 pairs



Example Food Items



...

Conclusion



- Investigation of cognitive structures and processes **is possible** without recourse to introspection and based on objective data
- Computational modeling helps to make theories more precise
- Using MDS, the confinement of computational models to artificial stimuli may be broken.

Thank you for your attention

- Looking forward to your comments & questions



